I mentioned in section 12.1.3 that the raw PSK cipher suites lack forward secrecy: if the PSK is compromised then all previously captured traffic can be easily decrypted. If confidentiality of data is important to your application and your devices can support a limited amount of public key cryptography, then you can opt for PSK cipher suites combined with ephemeral Diffie-Hellman key agreement to ensure forward secrecy. In these cipher suites, authentication of the client and server is still guaranteed by the PSK, but both parties generate random public-private key-pairs and swap the public keys during the handshake, as shown in figure 12.6. The output of a Diffie-Hellman key agreement between each side's ephemeral private key and the other party's ephemeral public key is then mixed into the derivation of the session keys. The magic of Diffie-Hellman ensures that the session keys can't be recovered by an attacker that observes the handshake messages, even if they later recover the PSK. The ephemeral private keys are scrubbed from memory as soon as the handshake completes.